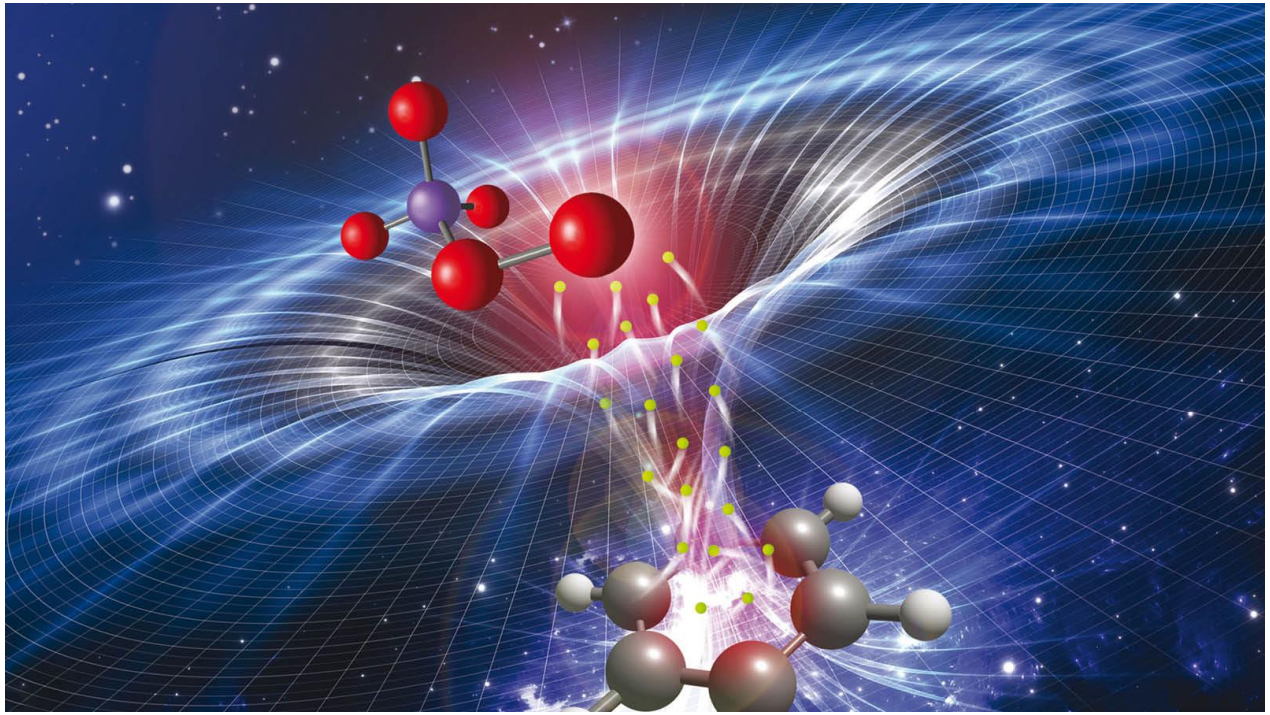
General Chemistry (C102)



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State of matter
Lecture 4

Liquefaction of gases

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Conditions of liquefaction of gases

- Many industrial processes require large quantities of liquid air, liquid ammonia, liquid carbon dioxide etc. The production of liquids from various gases is therefore an important commercial operation.
- There are different methods of liquefaction of gases, such as:
  - (i) based on the concept of critical temperature followed by the compression.
  - (ii) based on Joule-Thomson effect In the case of gases like NH_3, Cl_2, SO_2 and CO_2 whose T_c values are near and below the ordinary temperatures, they can be liquefied easily by increasing the pressure alone at their respective T_c values.
- Gases like H_2, O_2, N_2 and He have very low T_c values and hence Joule Thomson effect may be applied to bring in effective cooling.

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Joule Thomson Effect

- **Joule Thomson Effect:** When gas is allowed to expand adiabatically from a region of high pressure to a region of extremely low pressure, it is accompanied by cooling.”.

- During the high-pressure state, maximum intermolecular attraction force is occurred. During expansion, the molecules get far away from each other, as a result, they need to work against the intermolecular attraction force. The energy for this purpose is taken from the gas itself, so its temperature falls.

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Critical temperature and pressure

- Two important properties of gases are important in developing methods for their liquefaction: *critical temperature* and *critical pressure*. The critical temperature of a gas is defined as the temperature above which a gas cannot be liquefied. However, high pressure may be applied on the gas..
- The minimum pressure required to liquefy the gas at the critical temperature is called the *critical pressure*.
- For example, the critical temperature for carbon dioxide is 31°C. That means that no amount of pressure applied to a sample of carbon dioxide gas above 31°C will cause the gas to liquefy. At or below that temperature, the gas can be liquefied provided sufficient pressure is applied. The corresponding critical pressure for carbon dioxide at 31°C is 72.9 atmospheres. In other words, the application of a pressure of 72.9 atmospheres of pressure on a sample of carbon dioxide gas at 31°C will cause the gas to liquefy.

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Critical temperature and pressure

- Differences in critical temperatures among gases means that some gases are easier to liquefy than others.
- The critical temperature of carbon dioxide is high enough so that it can be liquefied relatively easily at or near room temperature.
- By comparison, ***the critical temperature of nitrogen gas is –147°C and that of helium is –267.7°C***.
- Liquefying gases such as nitrogen and helium present much greater difficulties than does the liquefaction of carbon dioxide.

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Methods of Liquefaction of Gases

- Faraday's method in which cooling is done with a freezing mixture
- Linde's method in which a compressed gas is released at a narrow jet (Joule-Thomson effect)
- Claude's method in which a gas is allowed to do mechanical work.

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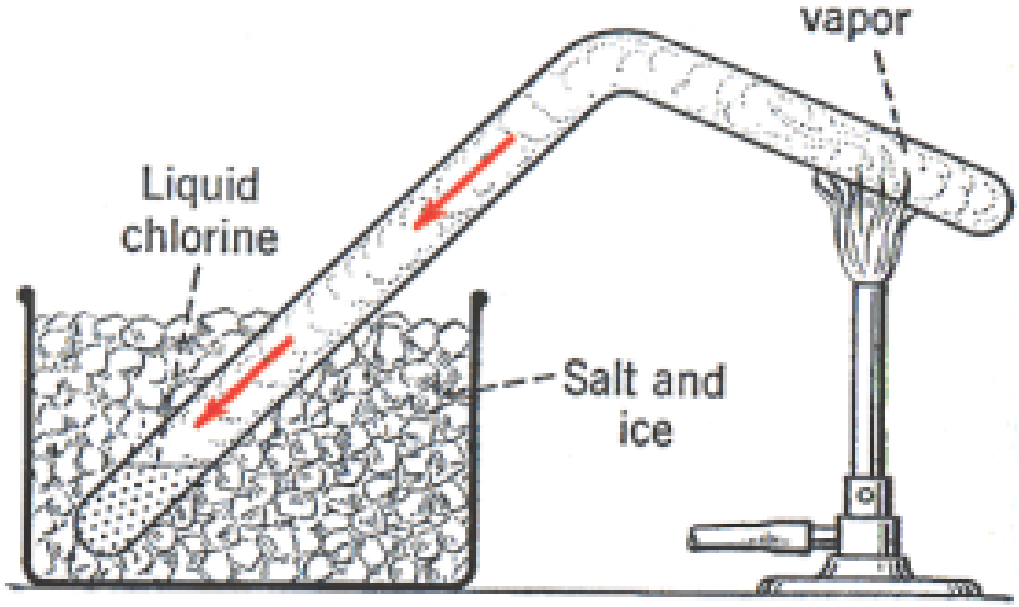
Faraday's Method

- **Faraday (1823)** used freezing mixtures of ice with various salts for external cooling of gases.
- The melting of ice and dissolution of salts both are endothermic processes.
- The temperature of the mixture is lowered up to a temperature when the solution becomes saturated.



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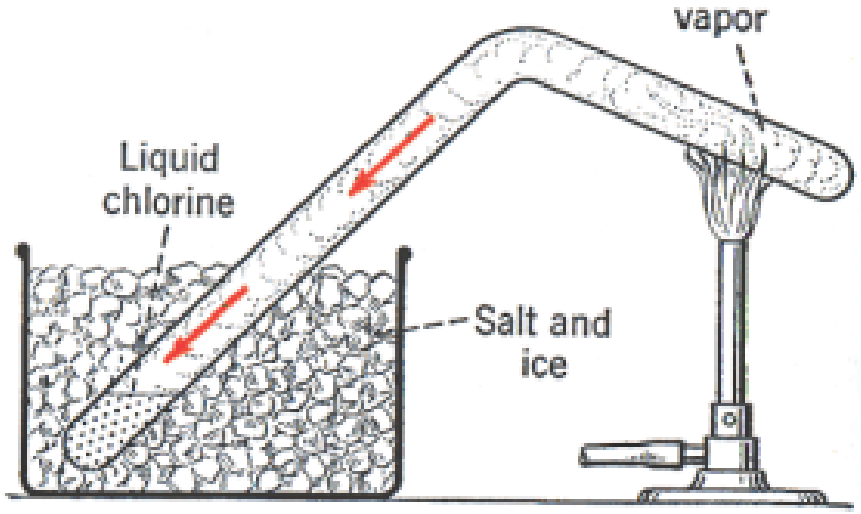
Faraday's Method

- Faraday succeeded in liquefying several gases such as **SO_2, CO_2, NO and Cl_2** by this method.
- He employed a V-shaped tube in one arm of which the gas was prepared. In the other arm, the gas was liquefied under *its own pressure*.
- The gases liquefied by this method had their critical temperature above or just below the ordinary atmospheric temperature.
- The other gases including **H_2, N_2 and O_2 (permanent gases)** having low critical points **could not be liquefied by Faraday's method.**



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Pictet Method (1877)

- **Pictet (1877)** succeeded in liquefying oxygen (O_2) for the first time and explained that they were no longer permanent gas, ***by applying the cascade method or a sequence of cooling***.
- When a gaseous liquid (liquefied gas) is rapidly evaporated, a low temperature can be achieved.
  - For example, when evaporating liquid sulfur dioxide, a temperature of (-56 ^{o}C) is enough to liquefy carbon dioxide.
  - When carbon dioxide is evaporated, it can reach a temperature of -130°C, which is sufficient to liquefy compressed oxygen.
  - Using the cascade method, it was able to reach lower temperatures, and the lowest temperature reached (-218°C) for oxygen by evaporating liquid oxygen.
  - But this temperature is much higher than the temperature for neon (T_C = - 228.71 ° C), helium (T_C = - 267.84 ° C) and hydrogen (T_C = - 239.9 ° C)

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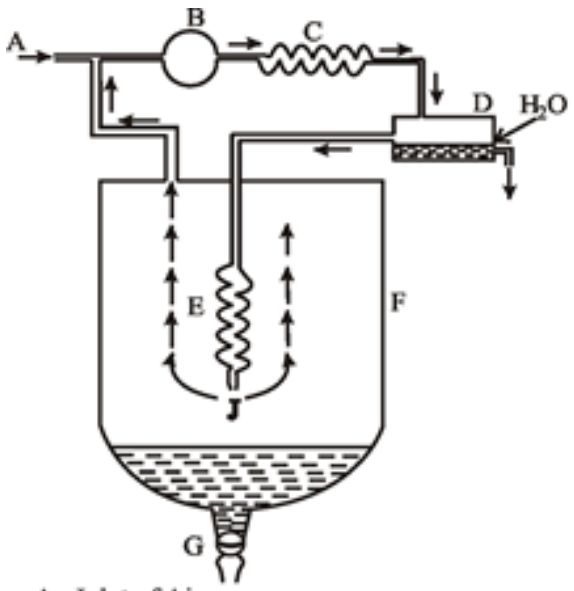
Hampson-Linde's method

- **Linde (1895)** used Joule Thomson effect as the basis for the liquefaction of gases.
- **When a compressed gas is allowed to expand into vacuum or a region of low pressure, it produces intense cooling.**
- In a compressed gas the molecules are very close and the attractions between them are appreciable. As the gas expands, the molecules move apart.
- In doing so, the intermolecular attraction must be overcome. The energy for it is taken from the gas itself which is thereby cooled and become liquid.



A - Inlet of Air
B - Compressor
C - Condensing Coils
D - Receiver for Water
E - Copper Spiral with a Jet J
F - Outer Chamber
G - Outlet for Liquefied Air

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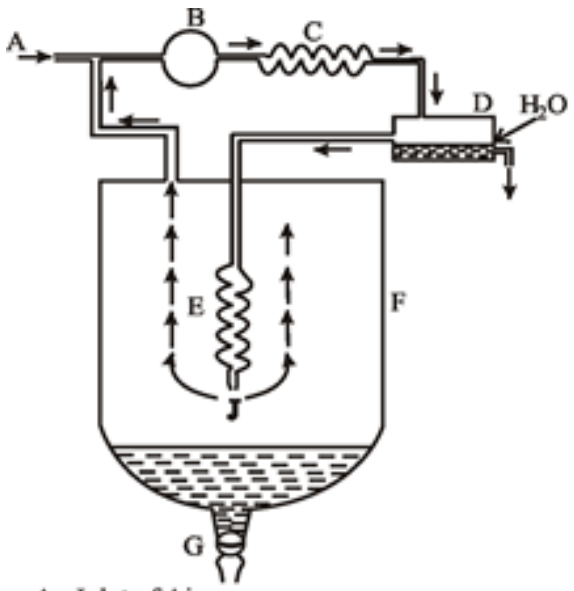
Hampson-Linde's method

- Linde used an apparatus worked on the above principle for the liquefaction of air. Pure dry air is compressed (B) to about 200 atmospheres.
- It is passed through a pipe (c) cooled by a refrigerating liquid such as ammonia. Here, the heat of compression is removed.
- The compressed air is then passed into a spiral pipe (J) with a jet at the lower end. The free expansion of air at the jet results in a considerable drop of temperature.



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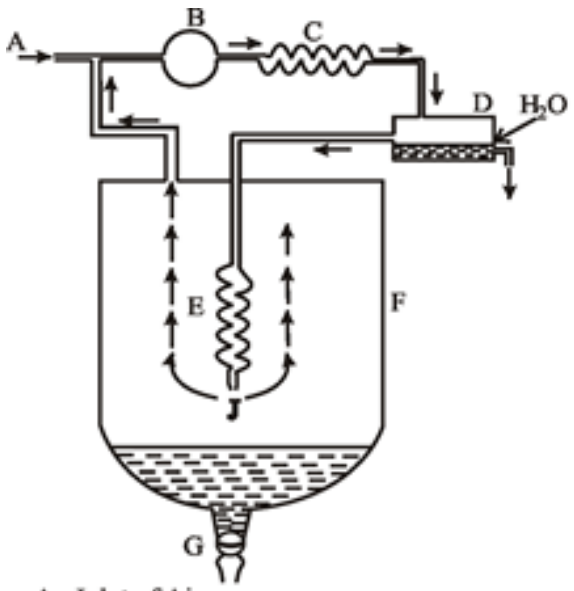
Hampson-Linde's method

- The cooled air which is now at about one atmosphere pressure passed up the expansion chamber (E).
- It further cools the incoming air of the spiral tube and returns to the compressor.
- By repeating the process of compression and expansion, a temperature low enough to liquefy air is reached.
- The liquefied air is collected at the bottom of the expansion chamber.



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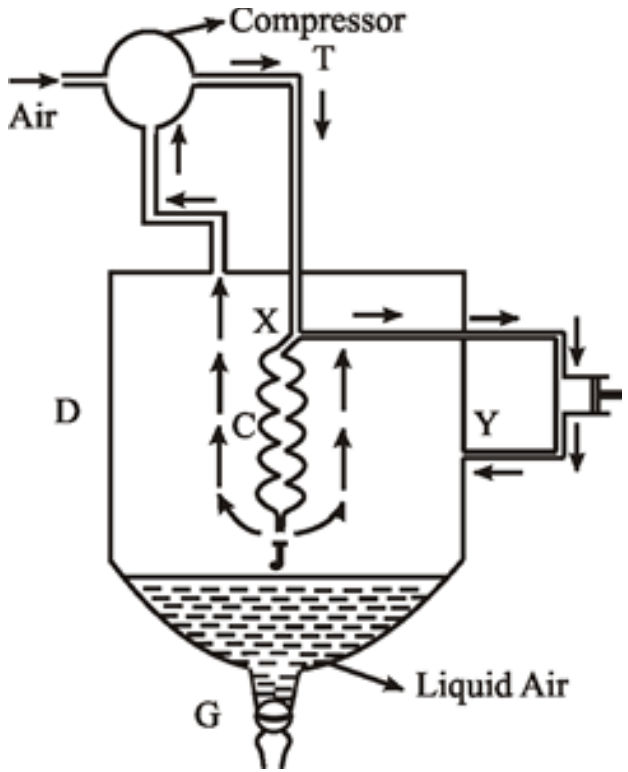
Cloud – Haylan's Method

- This method for liquefaction of gases is more efficient than that of Linde. Here, also the cooling is produced by free expansion of compressed gas. But in addition, the gas is made to do work by driving an engine (piston). The energy for it comes from the gas itself which cools.

- **Thus, in Claude's method the gas is cooled not only by overcoming the intermolecular forces but also by performance of work.** That is why, the cooling produced is greater than in Linde's method.



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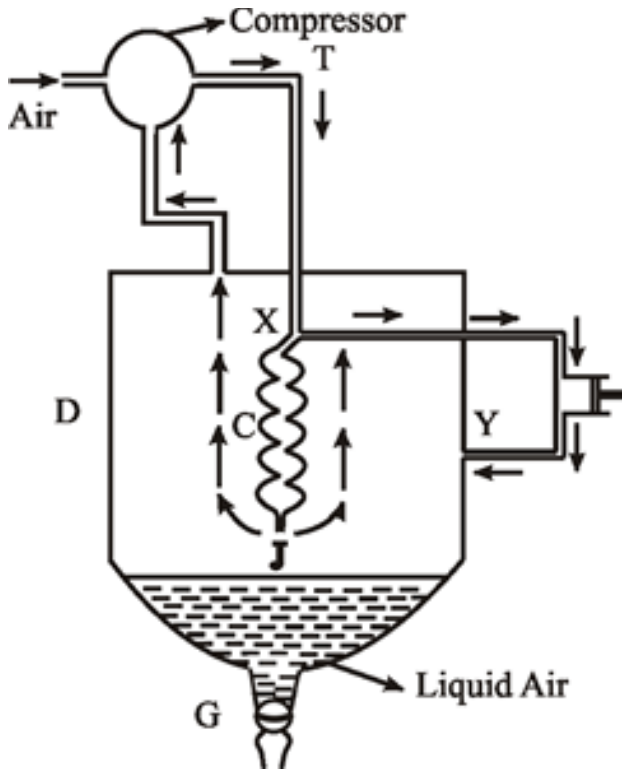
Cloud – Haylan's Method

- Pure dry air is compressed to about 200 atmospheres.
- It is led through a tube cooled by refrigerating liquid to remove any heat produced during the compression.
- The tube carrying the compressed air then enters the 'expansion chamber'. The tube bifurcates and a part of the air passes through the side-tube into the cylinder of an engine.
- Here it expands and pushes back the piston. Thus, the air does mechanical work whereby it cools.



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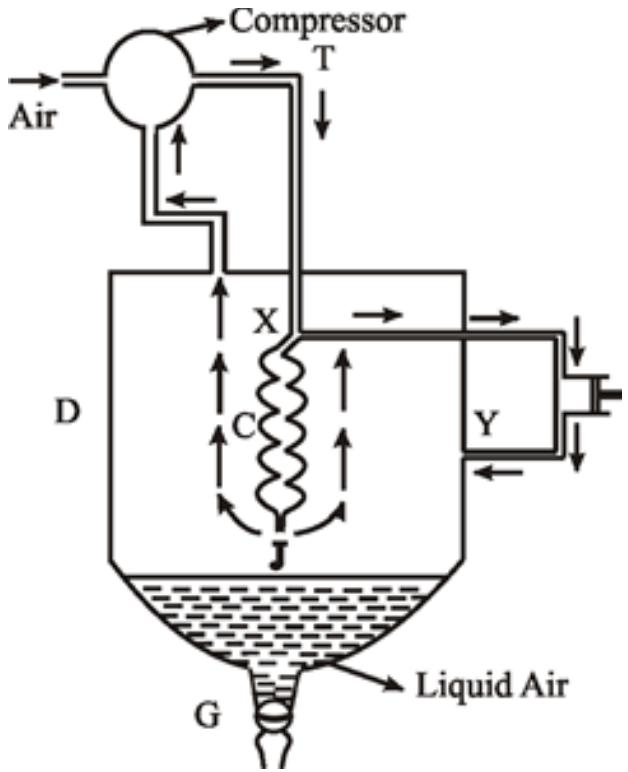
Cloud – Haylan's Method

- The air then enters the expansion chamber and cools the incoming compressed air through the spiral tube.
- The air undergoes further cooling by expansion at the jet and liquefies.
- The gas escaping liquefaction goes back to the compressor and the whole process is repeated.



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